Alexander Jacob

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EDUCATION

University of Southern California
Bachelor of Science in Electrical and Computer Engineering | GPA: 4.0 — Expected December 2029

EXPERIENCE

Rocket Propulsion Laboratory | Avionics Embedded Systems & PCB Design — 2026 - Present
University of Southern California, Los Angeles, CA
- Developed STM32 embedded C++ firmware for multi-node communication over half-duplex RS-485, integrating Mbed OS UART/GPIO interfaces and MAX485 transceiver control.
- Engineered a custom 4-byte PING/ACK protocol with device addressing, packet serialization, checksum validation, and receive-side command processing; verified protocol integrity through C++ unit tests.
- Modeled and simulated a 433 MHz helical antenna in Ansys, analyzing impedance matching, VSWR, gain, axial ratio, and radiation characteristics; investigated Yagi antenna SWR performance.
- Performed SMT assembly of an avionics camera-streaming PCB through solder-paste application, surface-mount component placement, and reflow soldering.

SLURM Lab | Assistant Researcher — Oct. 2026 - Present
University of Southern California, Los Angeles, CA
- Investigate DexMate robot teleoperation workflows to support tactile manipulation demonstrations for the FELT++ research project.
- Analyze experimental procedures for supervised robot control, teleoperation setup, and demonstration data collection.

Makers | Embedded Systems & PCB Design — 2026 - Present
University of Southern California, Los Angeles, CA
- Design software and PCB architecture for Soundtrack, a musical glove integrating five flex sensors, an ESP32, IMU, and Arduino Uno Q.
- Assess FluidSynth integration for C-based instrument synthesis and finger-controlled note and volume mapping.

PROJECTS

32-bit RISC-V Processor — 2026 - Present
SystemVerilog, RISC-V, Icarus Verilog, VS Code
- Designed a 32-bit RISC-V processor in SystemVerilog with an ALU, register file, instruction decoder, and program counter.
- Integrated datapath and control logic; wrote testbenches and used Icarus Verilog to validate instruction execution.

Mechanical Foosball — 2025 - 2026
VEXcode, VEX V5, Distance Sensors, 3D Printing
- Built a two-player game with 8 buttons, 6 motors, 2 distance sensors, 3D-printed players, and first-to-three scoring.
- Programmed bidirectional motor control, 125 mm goal detection, score tracking, and winner display in VEXcode.

Automated Connect 3 — 2024 - 2025
C++, Arduino Mega, Servos, CAD, 3D Printing
- Designed an accessible 6 x 4 game using an Arduino Mega, 8 micro servos, 6 column-selection buttons, and 24 3D-printed pieces.
- Wrote C++ control logic using a 2D array to validate moves, alternate turns, actuate servos, detect outcomes, and display game state.
- Modeled and 3D-printed 1,230 g of custom components; redesigned ball geometry after testing identified fit issues.

Joystick-Controlled 4-Direction LED PCB — 2026
KiCad, Arduino Uno, PCB Design, Circuit Prototyping
- Built and tested a joystick-controlled 4-direction LED circuit, then created a KiCad schematic and through-hole PCB with routed traces, a GND copper plane, labeled connectors, and ERC/DRC verification.

AWARDS

IBM Quantum Hackathon - Second Place — 2026
Qompute 2026, Track 2: Optimizing for People | LifelineQ
- Built LifelineQ to help choose which cooling shelters receive limited backup generators; compared quantum and classical computing results.

LEADERSHIP

President, Artificial Intelligence Club, Shaker High School — Sep. 2025 - Jun. 2026
- Led 20+ meetings on AI developments, applications, and ethics; managed officers and project leads, coordinated partnerships, and grew active membership to 30.

Eagle Scout, Troop 1277 — 2022 - 2026
- Led the design, construction, and installation of a 16-post Community StoryWalk connecting two local schools.

Front of House Team Member, Chick-fil-A — Apr. 2026 - Jul. 2026
- Processed orders, prepared beverages and desserts, and maintained dining areas in a high-volume environment.
- Collaborated with team members to maintain speed, accuracy, and guest service during peak periods.

SKILLS

Programming Languages: C/C++, Python, MATLAB, SystemVerilog
Hardware & Electronics: Circuit Design, PCB Design, Digital Logic Design, Embedded Systems, Computer Architecture
Engineering & Development: RF Engineering, Hardware Testing, Simulation & Modeling, Robotics, Design Verification
Software & Tools: Altium Designer, KiCad, Ansys, Arduino, Git/GitHub, Icarus Verilog